# Advances in Industrial and Systems Engineering

## Book 1: *Advances in Artificial Intelligence Applications in Industrial and Systems Engineering*

Over the last decade, industrial and systems engineering (ISE) has transformed with technological advancements like artificial intelligence (AI), digital manufacturing, and data analytics. *Advances in Artificial Intelligence Applications,* the first book in the *Advances in Industrial and Systems Engineering* series, edited by Gavriel Salvendy, Waldemar Karwowski, and Vincent Duffy, highlights the profound influence of AI in ISE, offering insights into its techniques, challenges, and applications across various ISE domains. Not only does it chart current AI trends and tools for effective integration, but it also raises pivotal ethical concerns, making it an essential guide for professionals integrating AI into system design and operation. This exciting new series presents the latest methodologies, tools, and real-world examples relevant to today's dynamic ISE landscape.

## Book 2: *Advances in Human-AI Collaboration*

*Advances in Human-AI Collaborations* is a natural follow-up to *Advances in AI Applications*, focusing on understanding how humans and AI systems work in tandem – a critical topic as we move toward an increasingly integrated tech landscape. This book will help ensure successful deployment and optimal benefits of AI applications in any setting. It will present pertinent material to provide an understanding of human-AI collaboration, opportunities, challenges, and applications of human-AI interaction

that can be adopted and applied in the industrial and systems engineering (ISE) field. Topics covered will include human intelligence vs. artificial intelligence, historical aspects of AI development, human-AI interaction fundamentals, training with AI support, AI and the future of work, and much more.

## Book 3: *Advances in Data Analytics and Applications*

*Advances in Data Analytics and Applications* is a comprehensive guide to understanding the relationship between data analytics and systems applications, a topic that is crucial for navigating the increasingly integrated technological landscape. Delving into the foundations, methodologies, and contemporary developments in data analytics, this guide addresses the opportunities and challenges inherent in its adoption across various industrial and systems engineering (ISE) domains. Covering topics ranging from forecasting and classification to machine learning-based analytics and ethical considerations, this book enables readers to harness the power of data analytics for enhanced productivity, quality, and customer satisfaction. Through its diverse range of topics and authoritative contributors, this book provides readers with the necessary skillset to navigate and leverage the transformative potential of data analytics in contemporary ISE practices.

## Book 4: *Advances in Modeling and Simulation*

*Advances in Modeling and Simulation* is a comprehensive guide to understanding the critical role of modeling and simulation in systems engineering, an essential area for navigating the increasingly complex technological landscape. Delving into the foundations, methodologies, and contemporary developments in modeling and simulation, this guide addresses the opportunities and challenges of its adoption across various industrial and systems engineering (ISE) domains. Covering topics ranging from data-driven discrete event simulation and AI modeling to virtual reality integration and federated learning, this book empowers readers to harness the power of modeling and simulation for enhanced productivity,

quality, and operational efficiency. Through its diverse range of topics and authoritative contributors, this book provides readers with the necessary skillset to navigate and leverage the transformative potential of modeling and simulation in contemporary ISE practices.

## Book 5: *Advances in Integrated Intelligent Systems Engineering*

*Advances in Integrated Intelligent Systems Engineering* offers a comprehensive exploration of the principles, methodologies, and applications shaping the future of systems engineering. It focuses on how artificial intelligence, machine learning, and model-based systems engineering are transforming the design and management of complex systems in industries such as healthcare, aerospace, and smart cities. Through real-world examples and case studies, the book provides practical insights into using intelligent systems to solve today's engineering challenges. With a strong emphasis on innovation, sustainability, and ethical considerations, this book focuses on practical applications, making it an essential resource for engineers, business leaders, and academics looking to excel in a hyper-connected, data-driven world.

## Book 6: *Advances in Human-Centric Industry 5.0*

*Advances in Human-Centric Industry 5.0* offers a comprehensive exploration of the next evolution in industrial transformation, where human collaboration, artificial intelligence, and sustainable practices redefine the future of work and production. This book defines the principles and methodologies that shape Industry 5.0, emphasizing the integration of AI, collaborative robotics, and digital transformation technologies across industries such as manufacturing, healthcare, and logistics. Through real-world case studies and expert insights, it provides a practical guide to navigating the challenges and opportunities of a more human-centered industrial paradigm. With a strong focus on innovation, sustainability, and ethical considerations, this book is an essential resource for engineers, business leaders, policymakers, and academics aiming to harness

the full potential of intelligent systems in an increasingly interconnected and technology-driven world.